Pulkit Katariya

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Education

Indian Institute of Information Technology and Management, Gwalior Gwalior, India
B.Tech in Mathematics and Scientific Computing *CGPA: 8.11/10 | Expected: 2028*

- **Relevant Coursework:** Data Structures & Algorithms, Design and Analysis of Algorithms, Object-Oriented Programming, Operating Systems, Software Engineering, Database Systems, Computer Networks, Linear Algebra, Probability & Statistics
- **Certification:** WorldQuant University – Applied Data Science Lab

Technical Skills

Languages: Python, Go, C++, C, SQL, JavaScript, TypeScript, Bash
Backend & Distributed: FastAPI, Strawberry GraphQL, Chi, REST APIs, SQLModel, SQLAlchemy, JWT/RBAC, Microservices
Frontend & Web: React, Next.js, HTML5, CSS3, Tailwind CSS, Responsive Web Design
Databases & Storage: PostgreSQL, MySQL, Redis (NoSQL), Apache Parquet, SQLite
Cloud & DevOps: Docker, Kubernetes, AWS (EC2, S3), GCP, CI/CD, GitHub Actions, Linux, Nginx, Git
AI/ML & Frameworks: PyTorch, XGBoost, LightGBM, Scikit-learn, Transformers (DistilBERT), MLflow, TreeSHAP

Open Source Contributions

Volcano – volcano-sh (CNCF Incubating Project)
Open Source Contributor | GitHub

- Authored comprehensive table-driven unit tests for all queue state transitions (`Open`, `Closed`, `Closing`, `Unknown`) in `pkg/controllers/queue/state/`, achieving 100% test coverage across the queue lifecycle for Volcano's HPC and AI batch scheduler.
- Engineered table-driven unit tests for all job controller state transitions in `pkg/controllers/job/state/`, achieving 100% statement coverage integrated into automated CI test suites to prevent regressions in distributed scheduling.

Projects

ACAD-ERP | GitHub | *FastAPI, Strawberry GraphQL, Next.js, React, PostgreSQL, Docker*

- Translated institutional business requirements into a modular ERP platform across 9 domains with role-gated access control for **Admin, Faculty, and Student** workflows.
- Implemented an authentication and RBAC system using JWT with HTTPBearer and Argon2 password hashing, including live role hydration for fine-grained, secure access control.
- Architected a unified **Strawberry GraphQL** API layer integrated with a responsive **Next.js / React** web client for real-time data queries and administrative workflows.

Harvest – YouTube Data Pipeline | GitHub | *Go, YouTube API, SQLite, Apache Parquet, AWS EC2, systemd*

- Architected a resilient 24/7 Go ingestion daemon deployed on **AWS EC2** with systemd supervision and structured logging, tracking video engagement under a strict **10,000 units/day** API quota budget across 3,500+ creator channels.
- Engineered a zero-quota discovery pipeline polling RSS/Atom feeds via an async HTTP worker pool, batching due observations (50 IDs/call) to achieve **0.02 quota units per snapshot**.
- Designed a 13-checkpoint non-uniform sampling state machine (0.5h–60h) with dual-stage comment harvesting, serializing sealed trajectories into **Snappy-compressed Apache Parquet** partitions.

Viral – Predictive ML & Analytics | GitHub | *Python, XGBoost, DistilBERT, FastAPI, MLflow, TreeSHAP*

- Formulated a leakage-free temporal prediction system across **65,881 videos** and **4.1M+ comments**, evaluating early engagement kinematics and sentiment signals ($t \leq 6h$) to forecast subsequent trending trajectories.
- Authored 7 research ablation experiments; engineered dual-stage comment drift using **DistilBERT** and **VADER** (+21.3% PR-AUC lift), with an optimized **XGBoost** model achieving **0.6206 PR-AUC** and **0.9784 ROC-AUC**.
- Integrated **TreeSHAP** for explainability, tracked metrics in **MLflow**, and deployed a high-throughput **FastAPI** REST service delivering sub-2ms low-latency inference.